

1. Sutherland 4.1
2. Sutherland 4.2

3. (AI Free)

Let $A \subset \mathbb{R}$ be a nonempty set that is bounded above. Let $m = \sup A$. Show that for every $\epsilon > 0$ there exists $a \in A$ with $m - \epsilon < a \leq m$.

4. Suppose (x_n) is a sequence converging to a limit x .
 - a) Show that (x_{n+1}) converges to the same limit.
 - b) Show that if $c \in \mathbb{R}$ then (cx_n) converges to cx .
5. Suppose $a \in \mathbb{R}$ and suppose $a^n \rightarrow L$. Use the previous problem to show that either $L = 0$ or $a = 1$ (in which case $L = 1$).
6. Suppose $0 < a < 1$. Use the monotone convergence theorem to show $a^n \rightarrow 0$. Hint: the previous problem could be helpful.
7.
 - a) Knowing that $\sqrt{2}$ is irrational, show that $q\sqrt{2}$ is irrational for all $q \in \mathbb{Q}$.
 - b) Sutherland 4.8. **Hint:** consider $a/\sqrt{2}$ and $b/\sqrt{2}$ as endpoints first.
8. Show directly from the definition that $x_n = (-1)^n$ is not Cauchy and conclude that it does not converge.
9. Sutherland 4.12 **Hint:** For simplicity, assume f and g are defined on all of $\mathbb{R}$, and take f to be really boring.